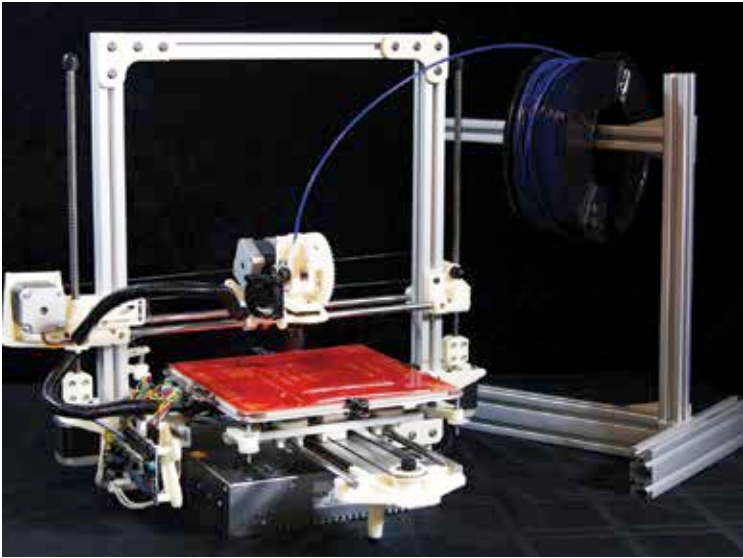




The Bukobot RepRap 3D printer

to the thickness of the layers. So, a 3D printer with a higher resolution can print thinner layers and is capable of greater accuracy.

3D printing drastically reduces production time compared to conventional methods and provides a higher degree of accuracy. And the exciting bit is that several projects and companies are making affordable 3D printers for home usage targeted at the DIY and maker communities. Some of the popular ones are:

- ♦ **RepRap:** One of the longest running projects, RepRap is a free and OSHW 3D printer, which is partially capable of replicating itself by printing many of its plastic parts. They also have the ability to print circuit boards and metal parts. The open source availability of printer designs has led to the emergence of a community and a vast number of derivative printers catering to various levels of accuracy and materials. As a result the cost of 3D printers has decreased dramatically in the last five years by almost 1/20th. Many companies are also selling parts to build your own RepRap 3D printer.
- ♦ **MakerBot:** MakerBot was one of the earliest projects, which was based off the RepRap project. The first 3D printers from MakerBot were built on top of the Arduino microcontrollers, and were sold as DIY kits. The